

POOJA GUTTAL

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EDUCATION

University of Maryland, Baltimore County

Master of Science in Computer Science

Aug 2024 – May 2026

CGPA: 3.82 / 4.0

Visvesvaraya Technological University

Bachelor of Engineering in Information Science

Aug 2018 - July 2022

CGPA: 3.51 / 4.0

SKILLS

Languages: Python, C, Java, JavaScript, TypeScript, HTML/CSS

Technologies: React, Spring Boot, Django, Node.js, Microservices, REST APIs

Databases: PostgreSQL, MySQL, MongoDB, Neo4j

Cloud: AWS (S3, EC2, Lambda, IAM), Google Cloud Platform (GCP)

AI/ML: NLP, Vector Search (Elastic, FAISS), RAG, LangChain, TensorFlow, PyTorch, NumPy, Scikit-learn, Fine-Tuning

Tools & Methodology: Git, GitHub, Bitbucket, Docker, Postman, Agile, CI/CD Pipelines

WORK EXPERIENCE

Python & AI Engineer (Co-op Intern)

Altumatim

Feb 2026 – May 2026

Remote, United States

- Engineered a production grade **LLM pipeline** on **Vertex AI (Gemini 2.5 Flash)** for legal filings, enabling 10-stage classification and entity extraction across client provided labels, replacing manual document review at scale.
- Designed a **structure-aware chunking algorithm for tabular documents**, outperforming LangChain recursive splitter by reducing context fragmentation by **35%** and **8x** faster processing across **RAG pipelines** - published on [arXiv](#)
- Implemented a speaker detection + transcription pipeline feature for legal deposition videos, achieving **22x real-time processing**, enabling **speaker-specific evidence retrieval** for legal investigation workflows.

Software Engineer

Persistent Systems

Aug 2022 – Aug 2024

Bangalore, India

- Led development of a **JavaScript (ES6+) browser automation system** for the internal Learning & Development team, generating quizzes from multimedia content, **reducing manual effort by 75%**.
- Built scalable full-stack solutions for **HealthEdge's** insurance claims platform, **improving workflow efficiency by 35%**.
- Re-engineered legacy frontend into responsive **React/Redux SPAs**, reducing page load time by **40%** and improving UI responsiveness and maintainability for an enterprise insurance portal.
- Developed scalable backend microservices using **Spring Boot with JWT-secured REST APIs**, enabling secure data access across distributed insurance claims systems for enterprise health plan clients.
- Optimized complex **SQL queries and database schemas (PostgreSQL)**, reducing query execution time **by 60%**.

Software Engineer Intern

L2MRail

Sep 2021 – Jan 2022

Bangalore, India

- Engineered features for a **Django/PostgreSQL-based rail wheel health monitoring system** and implemented real-time sensor data ingestion via asynchronous message queues.
- Optimized SQL queries, reducing latency by **50%** while handling **20,000+ daily sensor events**.
- Deployed and configured **AWS ECS (Linux)** infrastructure, ensuring service discovery and **99% uptime**.

PROJECTS

CBT-Guided Conversational AI for Mental Health Support | *Best MS-Led Research Award, CSEE 2026* – [GitHub](#)

- Built a **knowledge-guided NLP framework** using knowledge graph retrieval, cognitive model and Natural Language Inference filtering to enable clinically grounded LLM reasoning, applying responsible AI principles and safety constraints.
- Evaluated Chain-of-Thought prompting across 4 open-source models, achieving **66.7%** therapeutic reasoning adherence.

FactMed – AI-Driven Medical Chatbot | **Python, FAISS, BioBERT**– [GitHub](#)

- Developed a RAG-based medical chatbot using **FAISS vector search** and **BioBERT embeddings** with **47,000+ MedQuAD pairs**, grounding LLM responses in verified medical context and improving answer relevance over baseline LLM's.

LEADERSHIP

Guest Speaker, [Generative AI - SVIT Institute of Technology](#)